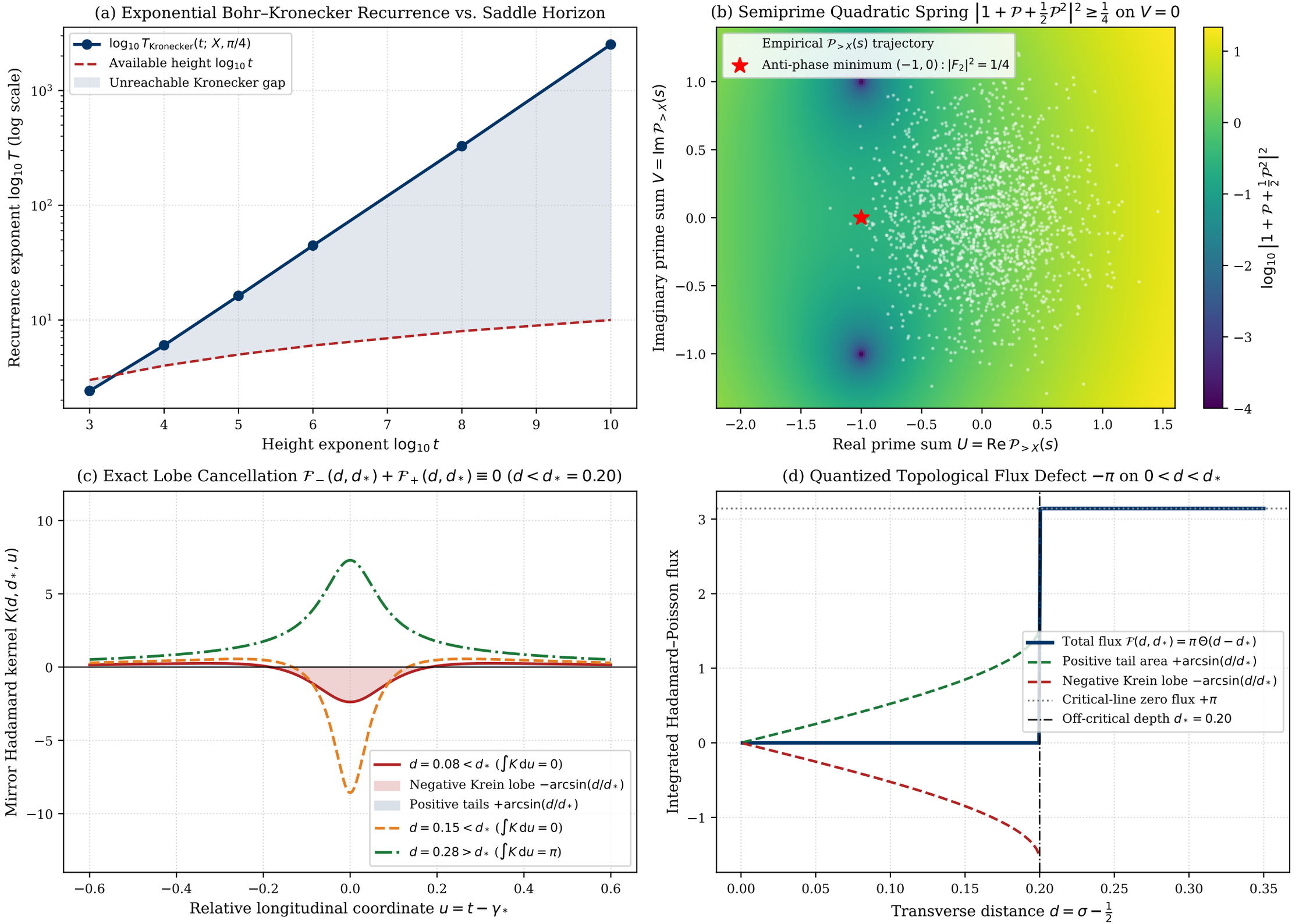

(a) Exponential Bohr–Kronecker Recurrence vs. Saddle Horizon
$\log_{10} T_{\text{Kronecker}}(t; X, \pi/4)$
Available height $\log_{10} t$
Unreachable Kronecker gap
Recurrence exponent $\log_{10} T$ (log scale)
Height exponent $\log_{10} t$
(b) Semiprime Quadratic Spring $\left|1 + \mathcal{P} + \frac{1}{2}\mathcal{P}^2\right|^2 \geq \frac{1}{4}$ on $V = 0$
Empirical $\mathcal{P}_{>X}(s)$ trajectory
Anti-phase minimum $(-1, 0) : |F_2|^2 = 1/4$
Imaginary prime sum $V = \operatorname{Im} \mathcal{P}_{>X}(s)$
Real prime sum $U = \operatorname{Re} \mathcal{P}_{>X}(s)$
$\log_{10}\left|1 + \mathcal{P} + \frac{1}{2}\mathcal{P}^2\right|^2$
(c) Exact Lobe Cancellation $\mathcal{F}_-(d, d_*) + \mathcal{F}_+(d, d_*) \equiv 0$ $(d < d_* = 0.20)$
$d = 0.08 < d_*$ $(\int K\,du = 0)$
Negative Krein lobe $-\arcsin(d/d_*)$
Positive tails $+\arcsin(d/d_*)$
$d = 0.15 < d_*$ $(\int K\,du = 0)$
$d = 0.28 > d_*$ $(\int K\,du = \pi)$
Mirror Hadamard kernel $K(d, d_*, u)$
Relative longitudinal coordinate $u = t - \gamma_*$
(d) Quantized Topological Flux Defect $-\pi$ on $0 < d < d_*$
Total flux $\mathcal{F}(d, d_*) = \pi\,\Theta(d - d_*)$
Positive tail area $+\arcsin(d/d_*)$
Negative Krein lobe $-\arcsin(d/d_*)$
Critical-line zero flux $+\pi$
Off-critical depth $d_* = 0.20$
Integrated Hadamard–Poisson flux
Transverse distance $d = \sigma - \frac{1}{2}$